

Forward Flux Sampling for Tropical Cyclogenesis in a Neural Weather Emulator

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Abstract

Tropical cyclone genesis is a rare atmospheric event whose probability varies by orders of magnitude across synoptic regimes, yet direct ensemble sampling cannot resolve this variability at operationally feasible ensemble sizes. We demonstrate that Forward Flux Sampling (FFS), a non-equilibrium rare-event technique from statistical mechanics, efficiently estimates genesis rates in a learned neural weather emulator (SDL-WXFormer, 1° resolution) without modifying model dynamics. FFS decomposes the rare $A \rightarrow B$ transition into a flux through an initial interface and a product of conditional crossing probabilities across four shooting interfaces, exploiting the natural parallelism of the shooting phase. Applied to 98 Atlantic basin initial conditions spanning August 21 – October 8, 2022, FFS resolves genesis rates spanning nearly three orders of magnitude (2.6×10^{-5} to 1.7×10^{-2} events day^{-1}), capturing a clear seasonal cycle invisible to 50-member direct ensemble forecasting. Internal validation — comparing FFS rates to independent direct-sampling rates from the same flux trajectories — yields a mean ratio of 1.03 ± 0.15 across all 98 initial conditions, confirming correct implementation of the conditional probability factorization. Computational enhancement factors range from $3\times$ (most active environment) to $140\times$ (most suppressed), with a geometric mean of $14\times$. Three case studies — Hurricane Earl (September 2), a peak-activity environment (September 9), and the Hurricane Ian precursor (September 22) — reveal physically distinct genesis bottlenecks: the rate-limiting step is initial tropical organization for Earl, uniformly high crossing probabilities for the active-period environment, and a hard barrier at the final intensification interface for the Ian precursor. These contrasting probability structures encode distinct atmospheric physics and demonstrate FFS as a diagnostic tool for mechanistic understanding of genesis pathways.

1 Introduction

Tropical cyclone genesis represents one of the most consequential rare events in atmospheric science. Approximately 10–15% of Atlantic tropical disturbances develop into named systems [Frank, 1970, Landsea, 1998], but discriminating developing from non-developing cases challenges deterministic and ensemble approaches alike [Titley and Elsberry, 2000, Halperin et al., 2013]. The computational

challenge is fundamental: for a genesis event with probability $p \sim 10^{-3}$ per day and target relative uncertainty of 20%, direct Monte Carlo estimation requires $\mathcal{O}(10^4)$ ensemble members per initial condition, far exceeding operational budgets.

Rare event sampling techniques address this by concentrating computational effort on transition pathways rather than waiting for rare events to occur spontaneously. The atmospheric rare-event literature has converged on Adaptive Multilevel Splitting (AMS) and variants as primary tools [Ragone et al., 2018, Webber et al., 2019, Finkel et al., 2024]. These methods efficiently generate ensembles of rare trajectory *realizations*. Forward Flux Sampling [Allen et al., 2006, 2009] occupies a complementary niche: it is designed specifically to estimate steady-state *rates* — expected transitions per unit time — with decomposed uncertainty that reveals which intensification stage is rate-limiting. For hurricane genesis, the operational question is not merely what a genesis event looks like but *how frequently* it occurs under a given atmospheric state. FFS delivers this directly.

FFS was developed for non-equilibrium steady-state (NESS) systems and requires no equilibrium assumptions: no free energy landscapes, detailed balance, or Boltzmann distributions. The method needs only well-defined metastable regions, measurable flux through interfaces, and Markovian dynamics [Allen et al., 2009]. The atmosphere is a canonical NESS system — driven by solar forcing, dissipated by radiation and friction — and satisfies all three conditions. Atmospheric rare events are dynamical transitions between regions of a strange attractor, not thermodynamic transitions over energy barriers [Touchette, 2009]. This makes FFS *more* defensible for atmospheric applications than equilibrium-based rare event techniques.

An important interpretive point: FFS genesis rates vary by three orders of magnitude across the 98 initial conditions studied here, reflecting strongly non-stationary background conditions across the hurricane season. The quantity k computed for each initial condition should therefore be understood as a *conditional genesis rate given the atmospheric state at that IC*, not a climatological steady-state rate. Each FFS simulation treats a 10–15 day forecast window anchored to a specific ERA5 state as a locally-stationary system, and the resulting k characterizes the genesis probability density in that particular atmospheric environment. This conditional framing — analogous to a conditional return period given present climate state [Finkel et al., 2024] — is the appropriate interpretation and motivates the full seasonal sampling across 98 ICs.

Machine learning weather models [Lam and Coauthors, 2023, Bi and Coauthors, 2023, Pathak and Coauthors, 2022] offer a natural platform for rare event sampling: they are $\sim 1000\times$ faster than physics-based models, natively generate ensemble spread through learned stochasticity, and inherit the attractor structure of the training data. The SDL-WXFormer [Schreck and Coauthors, 2024] extends this with stochastic decomposition layers calibrated to ERA5 ensemble statistics, making it suitable for the large trajectory counts required by FFS.

This paper demonstrates FFS for Atlantic tropical cyclogenesis in SDL-WXFormer, applied to 98 initial conditions from the 2022 hurricane season. We report (1) a three-order-of-magnitude seasonal cycle in genesis rates; (2) internal validation showing direct-sampling and FFS rates agree within 3% on average; (3) case studies revealing physically distinct rate-limiting steps across atmo-

spheric regimes; and (4) a Cyclone Phase Space contamination screen ensuring sampled pathways correspond to tropical rather than extratropical development.

2 Methods

2.1 Forward Flux Sampling

FFS estimates the transition rate from metastable state A (unorganized disturbances, $Q > \lambda_A$) to state B (organized tropical cyclone, $Q \leq \lambda_B$) by factorizing:

$$k_{A \rightarrow B}^{\text{FFS}} = \Phi_{A,0} \times \prod_{i=1}^n P(\lambda_i | \lambda_{i-1}) \quad (1)$$

where $\Phi_{A,0} = N_{\text{cross}}/T_{\text{total}}$ is the flux of trajectories crossing the first interface λ_0 per unit time, and $P(\lambda_i | \lambda_{i-1})$ is the conditional probability that a trajectory at λ_{i-1} reaches λ_i before returning to A . The order parameter $Q(\mathbf{x})$ is the minimum sea-level pressure (MSLP) within the Atlantic basin domain $\Omega = [10^\circ\text{N}, 40^\circ\text{N}] \times [100^\circ\text{W}, 20^\circ\text{W}]$.

State definitions and interfaces. State A : $Q > \lambda_A = 1008$ hPa (unorganized synoptic disturbances). State B : $Q \leq \lambda_B = 975$ hPa (tropical storm to minimal hurricane intensity). The 975 hPa threshold is calibrated to the SDL-WXFormer’s 1° horizontal resolution: at this resolution the model reliably represents mesoscale circulation and warm-core development through 975 hPa but cannot be expected to faithfully reproduce intensity fluctuations at ≤ 960 hPa characteristic of major hurricanes, which require convection-permitting resolution. The flux interface and four shooting interfaces are:

$$\lambda_0 = 1000.0, \quad \lambda_1 = 987.2, \quad \lambda_2 = 984.7, \quad \lambda_3 = 981.2, \quad \lambda_4 = \lambda_B = 975.0 \quad [\text{hPa}]$$

satisfying $\lambda_A > \lambda_0 > \lambda_1 > \dots > \lambda_4 = \lambda_B$.

Interface placement. Shooting interfaces are chosen using the optimal- $1/e$ rule of Allen et al. [2009]: the next interface λ_{i+1} is placed at the p^* -quantile of the MSLP distribution reached by trajectories launched from λ_i , where $p^* = 1/e \approx 0.368$. This placement minimises variance in the estimated rate and targets conditional crossing probabilities near $1/e$, maximising statistical efficiency [Allen et al., 2009]. In practice, interfaces were optimized from long trial runs and then rounded to physically interpretable pressure values; the resulting P_i values (Table 1) cluster between 0.10 and 0.91 across all ICs, with a majority in the well-conditioned range 0.25–0.75 as shown in Figure 5.

Flux estimation. For each initial condition we generate trajectories from the ERA5 state perturbed by SDL-WXFormer stochastic layers, advancing 6-hourly for up to 15 days. Decorrelated λ_0 crossings (requiring return to state A between consecutive crossings) are accumulated to $N_{\text{cross}} = 2000$ per initial condition, yielding Poisson uncertainty $\sigma_\Phi/\Phi \approx 2.2\%$ — well below across-IC variability. State- B arrivals during flux generation provide an independent direct-sampling rate

$k^{\text{direct}} = N_{\text{direct}}/T_{\text{total}}$ that serves as internal validation.

Shooting. Configurations saved at λ_0 seed shooting trajectories toward each successive interface. For each interface λ_{i-1} , trajectories are launched until 2000 successful crossings of λ_i are accumulated. Success probability $P(\lambda_i|\lambda_{i-1}) = N_{\text{success}}/N_{\text{attempts}}$ with binomial uncertainty. Statistical uncertainty on the full rate (Eq. 1) propagates from each factor:

$$\frac{\sigma_k}{k} = \sqrt{\left(\frac{\sigma_\Phi}{\Phi}\right)^2 + \sum_{i=1}^n \left(\frac{\sigma_{P_i}}{P_i}\right)^2} \quad (2)$$

2.2 SDL-WXFormer and Experimental Setup

SDL-WXFormer [Schreck and Coauthors, 2024] is a vision transformer trained on ERA5 reanalysis (1979–2021) at 1° resolution, 6-hour timesteps, predicting 69 pressure-level and 8 surface variables. Stochastic decomposition layers inject learned, state-dependent noise enabling calibrated ensemble generation. Each 10-day forecast requires ≈ 1.5 GPU-minutes on NVIDIA A100.

We select 98 initial conditions from ERA5 at 00 and 12 UTC spanning August 21 – October 8, 2022, verifying $Q(\mathbf{x}_0) > 1008$ hPa for each. Three initial conditions receive detailed case study treatment:

- **2022-09-02T00Z (Earl):** ~ 24 hours before Hurricane Earl’s official tropical-storm genesis on September 3.
- **2022-09-09T12Z (active):** highest observed FFS genesis rate in the dataset, within the Fiona precursor environment (~ 5 days before Fiona’s formation).
- **2022-09-22T00Z (Ian precursor):** 24 hours before Hurricane Ian’s NHC genesis designation on September 23.

2.3 Feature Tracking and CPS Contamination Screen

Flux mode (multi-storm tracking). During the flux phase, the order parameter Q is the global minimum MSLP over the basin domain Ω , which may correspond to any co-existing tropical system. To track this minimum coherently across timesteps, we maintain a running list of all active vortex candidates. At each 6-hour step, Gaussian smoothing ($\sigma = 1.5$ grid-points) is applied to the MSLP field and local minima below 1010 hPa are detected. Existing tracks are updated by nearest-neighbour matching within 12° ; unmatched minima below 12°N over land are suppressed (terrain artefacts); remaining unmatched minima seed new tracks. A track is retained for up to two consecutive timesteps of failed matching before being dropped. This scheme simultaneously follows multiple co-existing systems and switches the designated “minimum” when a new system deepens below the current one, preserving a well-defined monotone order parameter throughout the flux phase.

Shoot mode (single-vortex tracking). Once a trajectory has been saved at λ_0 and begins the shooting phase, the target vortex is the one that triggered the λ_0 crossing. At each timestep,

the MSLP minimum is sought within 12° of the vortex's last known position. This single-vortex constraint prevents Q from jumping to a stronger but unrelated system, ensuring that the shooting trajectory measures the intensification of the specific disturbance seeded at λ_0 .

CPS contamination screen. A simplified Cyclone Phase Space [Hart, 2003] filter screens contamination from two sources at 1° resolution: (1) spurious MSLP minima over high terrain from hydrostatic extrapolation errors, and (2) mid-latitude baroclinic cyclones entering the domain boundary. CPS verification is applied at λ_0 crossings and when storms exceed 50°N or -10°W , rejecting clearly extratropical systems ($-V_T^L > 0$ or $-V_T^U > 0$). We note that formal CPS thresholds are not validated for 1° model output; the filter is a practical contamination screen rather than a rigorous warm-core classifier.

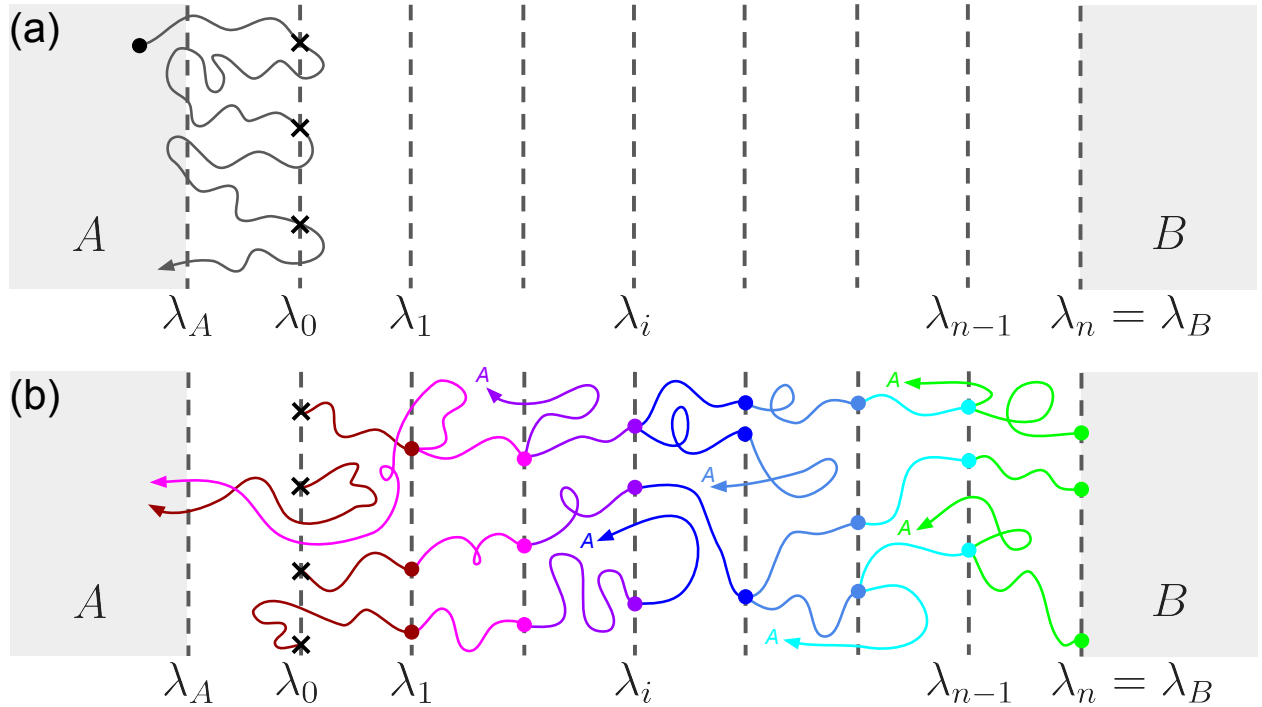


Figure 1: Schematic of the Forward Flux Sampling algorithm (after Allen et al. 2006). *Left panel (flux phase)*: Trajectories from state A (gray, $Q > 1008$ hPa) evolve forward in time; decorrelated crossings of λ_0 (orange circles) are saved. Trajectories reaching state B directly are tallied for direct-sampling rate validation. *Right panel (shooting phase)*: Trajectories launched from saved λ_i configurations either return to A (purple; failed) or advance to λ_{i+1} (dark blue; success), yielding $P(\lambda_{i+1}|\lambda_i)$. The genesis rate is $k = \Phi_{A,0} \prod_i P(\lambda_{i+1}|\lambda_i)$.

3 Results

3.1 Seasonal Genesis Rates and Internal Validation

Figure 2 shows FFS genesis rates across all 98 initial conditions. The full seasonal cycle spans nearly three orders of magnitude, from $2.6 \times 10^{-5} \text{ day}^{-1}$ during suppressed October conditions

to $1.7 \times 10^{-2} \text{ day}^{-1}$ during the peak-activity period around September 9–10 (consistent with the Fiona formation environment). This dynamic range is entirely unresolvable by direct 50-member ensemble sampling: the ECMWF IFS 50-member ensemble, analyzed over the same 98 initial conditions, reaches the minimum MSLP threshold in 25.4% of members on average. We note that this IFS statistic cannot be directly interpreted as a genesis rate: IFS uses 0.25° physics coarsened to 1.5° output, does not apply the decorrelated-crossing requirement that defines λ_0 flux, and includes extratropical systems without a CPS filter. We include this figure only as a contextual upper bound indicating that strong MSLP deepening occurs at a frequency broadly consistent with SDL-WXFormer, while acknowledging that day-to-day variability and the rate magnitude itself are meaningful only for the FFS estimates.

The primary internal validation is the agreement between k^{FFS} and the independent direct-sampling rate k^{direct} (Figure 2, orange triangles; scatter comparison in Figure 4). These estimates are derived from entirely separate components of the calculation: k^{FFS} is a product of flux and shooting probabilities, while k^{direct} counts raw state- B arrivals during flux generation. Across all 98 initial conditions, the ratio $k^{\text{FFS}}/k^{\text{direct}} = 1.03 \pm 0.15$ (mean $\pm$ std; individual ratios range 0.71–1.52). Agreement within 3% on average provides strong evidence that the conditional probability factorization is correctly implemented and that the interface ensemble sizes are adequate.

Computational enhancement factors range from $3\times$ (most active IC) to $140\times$ (most suppressed), with geometric mean $14\times$ and arithmetic mean $26\times$; the right-skewed distribution (geometric $\ll$ arithmetic mean) indicates a log-normal speedup distribution, consistent with the FFS rate itself spanning orders of magnitude. The geometric mean is the more appropriate summary statistic for log-normal quantities.

3.2 Case Studies: Distinct Genesis Bottlenecks

Table 1 presents FFS statistics for the three case-study initial conditions. The most striking feature is the variable position of the rate-limiting step, encoded in which conditional probability is smallest.

Table 1: FFS statistics for three case-study initial conditions. $\Phi_{A,0}$: flux rate. P_i : conditional crossing probability at interface λ_i . k^{FFS} : full genesis rate (Eq. 1). k^{direct} : independent direct-sampling rate. Ratio: $k^{\text{FFS}}/k^{\text{direct}}$. Speedup: enhancement over direct sampling for equivalent 20% rate uncertainty. Bottom row: statistics across all 98 ICs.

IC	$\Phi_{A,0}$ (day^{-1})	P_1	P_2	P_3	P_4	k^{FFS} (day^{-1})	k^{direct} (day^{-1})	Ratio	Speedup
Earl (09-02T00Z)	0.0837	0.258	0.679	0.596	0.359	3.13×10^{-3}	3.61×10^{-3}	0.87	$10\times$
Active (09-09T12Z)	0.1002	0.726	0.906	0.910	0.276	1.66×10^{-2}	1.37×10^{-2}	1.21	$3\times$
Ian pre (09-22T00Z)	0.1247	0.663	0.893	0.302	0.102	2.28×10^{-3}	2.04×10^{-3}	1.12	$19\times$
All-98 mean $\pm$ SD	0.065 ± 0.034	0.278 ± 0.154	0.674 ± 0.086	0.569 ± 0.107	0.356 ± 0.093	2.6×10^{-3}	2.5×10^{-3}	1.03 ± 0.15	$26 \pm 27\times$

For the **Earl IC**, $P_1 = 0.258$ is the dominant bottleneck: fewer than 1 in 4 trajectories crossing 1000 hPa organize to 987 hPa in this environment. This is physically consistent with the pre-genesis atmospheric state ~ 24 hours before Earl’s classification, when the system had established a coherent low-level circulation but had not yet achieved the sustained convective organization

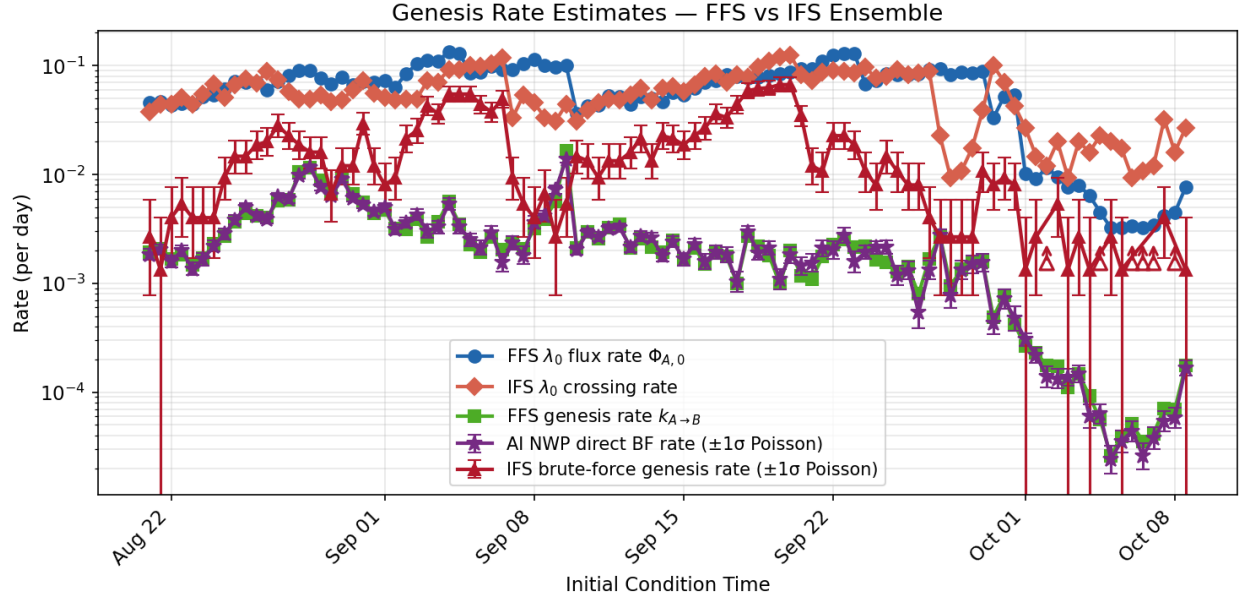


Figure 2: FFS genesis rates across all 98 initial conditions (August 21 – October 8, 2022) on a logarithmic scale. *Blue*: λ_0 flux rate $\Phi_{A,0}$. *Orange*: IFS λ_0 crossing rate (contextual reference; not directly comparable to FFS due to different resolution and no CPS filter). *Green*: Full FFS genesis rate k^{FFS} (flux $\times$ product of conditional probabilities). *Purple*: Direct-sampling rate k^{direct} from state- B arrivals during the flux phase ($\pm 1\sigma$ Poisson). *Red*: IFS brute-force genesis rate ($\pm 1\sigma$ Poisson; upward arrows denote upper bounds where zero events observed). Agreement between green and purple (mean ratio 1.03 ± 0.15) validates the FFS method. The FFS rate resolves a seasonal cycle spanning nearly three orders of magnitude invisible to direct 50-member ensemble sampling.

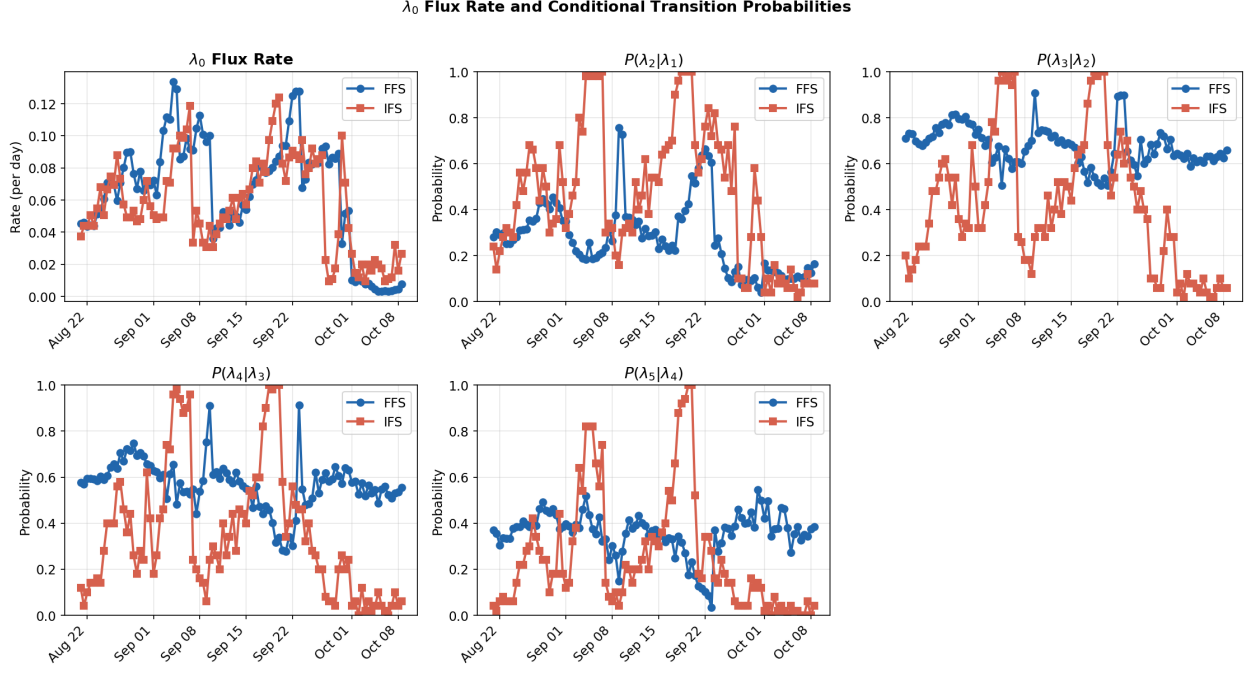


Figure 3: Seasonal evolution of FFS conditional crossing probabilities $P(\lambda_i|\lambda_{i-1})$ for all 98 initial conditions (August 21 – October 8, 2022). The first panel shows the λ_0 flux rate; subsequent panels show each P_i between successive shooting interfaces. The dashed line marks the all-season mean. P_1 (initial organization, $\lambda_1 \rightarrow \lambda_2$) varies most strongly with the seasonal cycle, with a pronounced peak in early September coinciding with the Fiona formation environment. P_4 (final intensification, $\lambda_4 \rightarrow B$) is the most uniformly suppressed probability, limiting the overall genesis rate even in active periods. The three case-study ICs (Earl 09-02, active 09-09, Ian 09-22) are indicated by vertical markers.

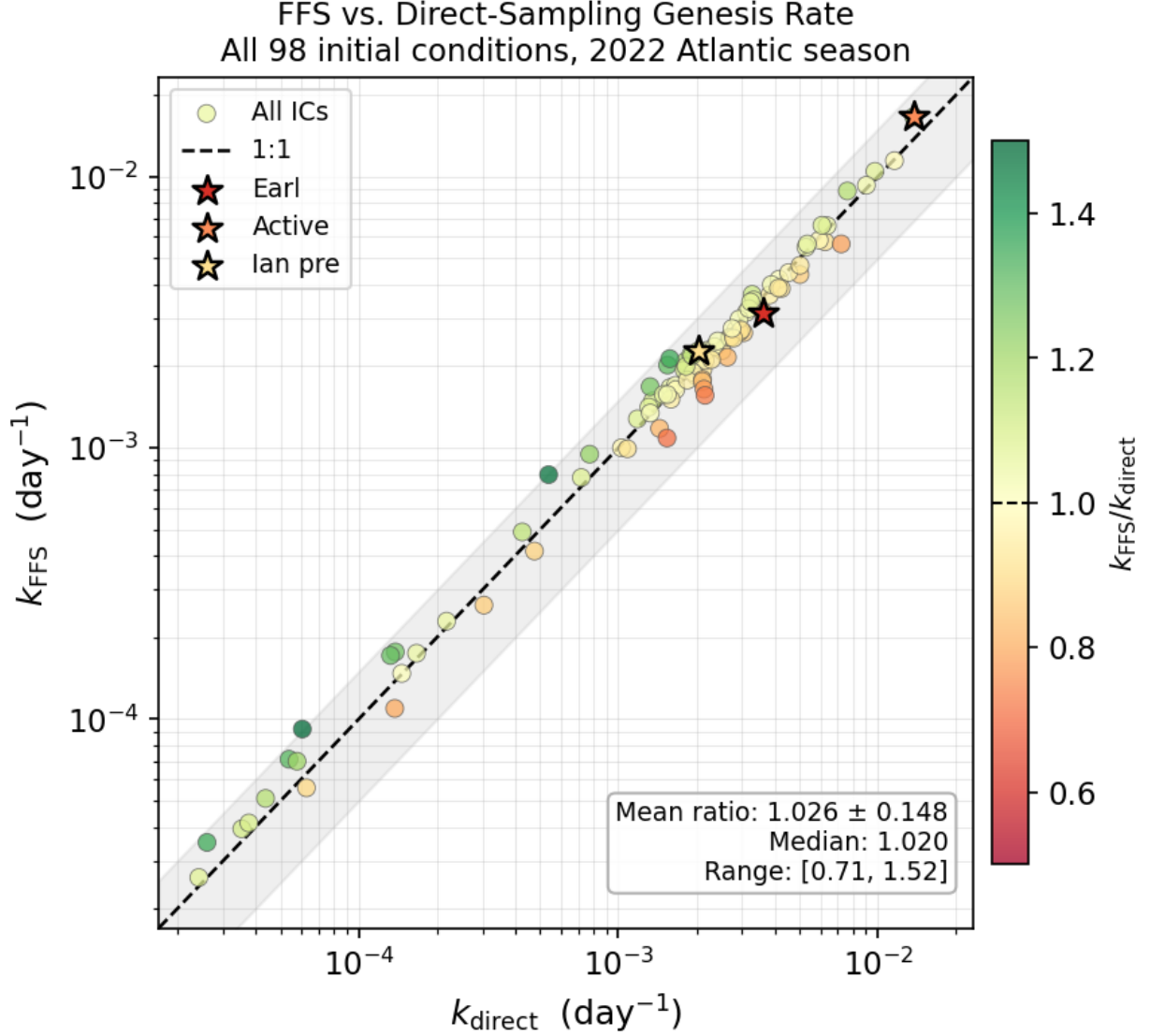


Figure 4: Log-log scatter of FFS genesis rate k^{FFS} versus independent direct-sampling rate k^{direct} for all 98 initial conditions. Points are coloured by the ratio $k^{\text{FFS}}/k^{\text{direct}}$: green indicates near-unity agreement; red/blue indicate over- or under-estimation. The black dashed line is the 1:1 reference; shaded band spans $\pm 50\%$. Three case-study ICs are marked with stars (Earl: red; active: orange; Ian pre: yellow). The tight clustering along the 1:1 line across three orders of magnitude confirms correct implementation of the conditional probability factorization. Inset statistics: mean ratio 1.03 ± 0.15 , median 1.00, range [0.71, 1.52].

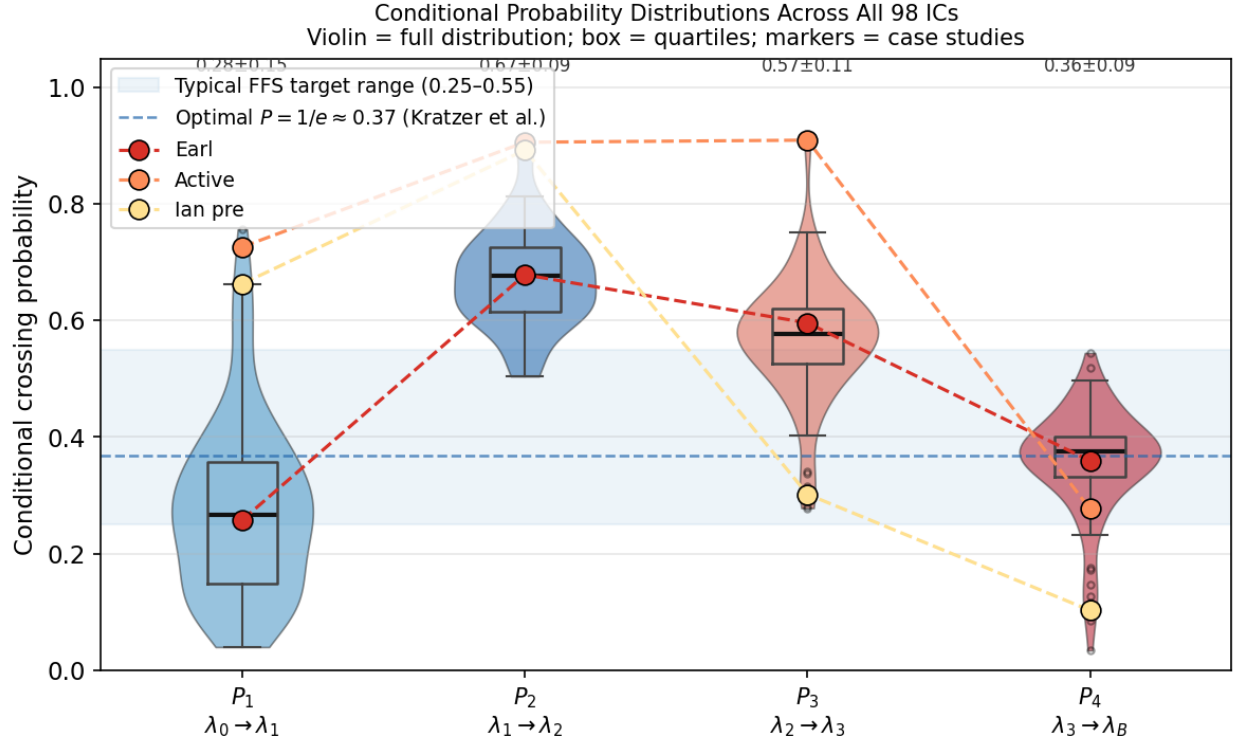


Figure 5: Distribution of conditional crossing probabilities P_i across all 98 initial conditions. Each violin shows the full distribution; boxes show quartiles; coloured lines connect the three case-study ICs. The horizontal dashed line marks the optimal $P = 1/e \approx 0.37$ (Kratzer et al.); the blue band spans the target range 0.25–0.55. Most ICs have P_i within the well-conditioned range, confirming that interface placement produces efficient variance reduction. The spread of P_1 (initial organization) is larger than that of P_2 – P_3 , reflecting the dominant role of large-scale atmospheric variability in setting the first intensification barrier. Mean $\pm$ SD for each interface are annotated.

needed to cross the first intensification threshold.

For the **active-period IC** (September 9), all four conditional probabilities are elevated (≥ 0.28), and the product $\prod P_i = 0.166$ is $\sim 4\text{--}9\times$ larger than the other two cases. The dominant bottleneck shifts to $P_4 = 0.276$ (final intensification to 975 hPa), reflecting an environment that readily organizes disturbances but where crossing the hurricane-threshold step remains non-trivial. The FFS rate ($1.66 \times 10^{-2} \text{ day}^{-1}$) is high enough that direct sampling is also feasible (speedup $3\times$), providing a strong cross-validation: both methods agree to within 21% for this most-active IC.

The **Ian precursor IC** presents the most physically interesting pattern. $P_1 = 0.663$ and $P_2 = 0.893$ indicate that the environment readily supports initial organization through 985 hPa, consistent with the warm, moist, low-shear conditions present over the Caribbean and western Atlantic on September 22. But $P_3 = 0.302$ and $P_4 = 0.102$ reveal a hard barrier in the final intensification steps (981 hPa to 975 hPa). This is physically consistent with Ian’s trajectory: the storm was still organizing over the Caribbean on September 22–23, entering the Gulf of Mexico’s warm core environment only after September 25, when conditions supported rapid intensification. The FFS framework identifies the final-stage intensification barrier *one day before* Ian’s official genesis — a diagnostic that direct ensemble sampling cannot resolve.

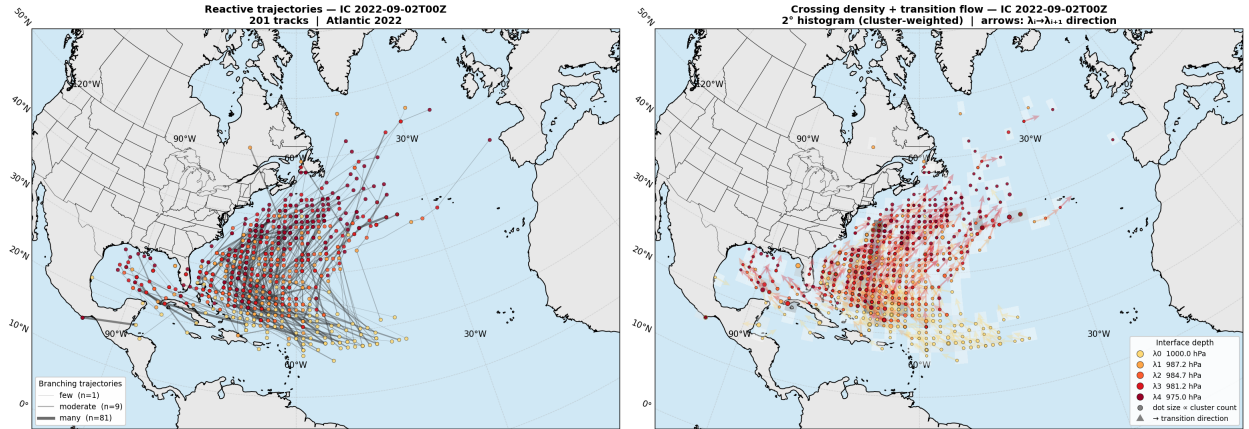


Figure 6: Reactive genesis trajectories for the Earl IC (2022-09-02T00Z). *Left*: Individual tracks that cross from λ_0 through to state B , color-coded by branching factor (cluster weight); the 201 reactive pathways sample a broad region spanning the tropical Atlantic, Caribbean, and western subtropical Atlantic. *Right*: Cluster-weighted crossing density (2° histogram) and mean $\lambda_n \rightarrow \lambda_{n+1}$ transition direction (arrows), revealing the geographic regions most active in the intensification cascade for this environment. CPS-rejected (extratropical) trajectories are excluded from both panels.

3.3 FFS Shooting Trees

Figure 7 shows representative FFS shooting trees for each of the three case-study ICs. Each tree traces the branching genealogy of a single λ_0 seed: the star marks the seed location, and successive

interface crossings fan out as the ensemble of reactive pathways progresses toward state B . The geographic spread of λ_4 descendants reflects the diversity of intensification pathways sampled by the FFS from a single starting configuration.

Importantly, the IC date labels (09-02, 09-09, 09-22) are *initialisation* times, not genesis dates. Each FFS simulation runs 15 days forward from its IC; each node in the tree represents the storm feature at a different future lead time within that window. A single seed at λ_0 therefore spawns descendants that span a range of calendar dates, geographic positions, and atmospheric states — all sampled from one background IC.

The Earl tree (left) originates near 31°N in the western Atlantic and fans northeastward, sampling the open Atlantic development region. The Fiona precursor tree (centre) seeds near 19°N in the open Atlantic and fans northwestward toward the Caribbean and Bahamas; Fiona was designated a tropical storm on September 14 — five days into this simulation window. The Ian precursor tree (right) seeds near 18°N in the Caribbean and fans northwestward into the Gulf of Mexico, directly anticipating Ian’s observed path.

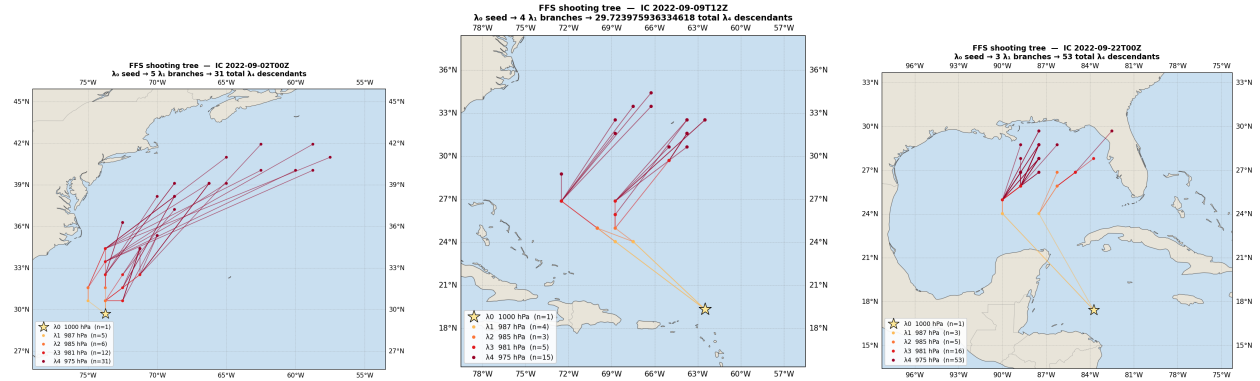


Figure 7: FFS shooting trees for the three case-study ICs (initialisation dates shown; each simulation runs 15 days forward, so each node represents a different future lead time within that window, not the IC date itself). Each panel shows one λ_0 seed (star) and the branching cascade of interface crossings to λ_4 ; marker colours indicate interface level. *Left*: Earl IC (init. 2022-09-02T00Z) — seed near 31°N , 72°W ; trajectories fan northeastward into the open Atlantic. *Centre*: Fiona precursor IC (init. 2022-09-09T12Z) — seed near 19°N , 63°W ; Fiona was designated on day 5 of this window; trajectories track northwestward consistent with its subsequent path. *Right*: Ian precursor IC (init. 2022-09-22T00Z) — seed near 18°N , 84°W ; trajectories fan northwestward into the Gulf of Mexico, anticipating Ian’s observed track.

3.4 Committor Curves and Spatial Genesis Fields

Figure 8 shows the committor curve $p_B(\lambda_i)$ — the probability of eventually reaching state B given arrival at interface λ_i — averaged across all 98 initial conditions for both the FFS (AI model) and IFS (brute-force ensemble). The all-season FFS curve shows a near-linear decline with the steepest drop between the first and second shooting interfaces ($\lambda_1 \rightarrow \lambda_2$), consistent with the seasonal mean of P_1 being the lowest of the four conditional probabilities (Table 1). The FFS and IFS curves agree

in overall shape, with the FFS predicting systematically higher committor values at each interface, reflecting the higher genesis rate of the AI model relative to IFS at $0.25^\circ/1.5^\circ$.

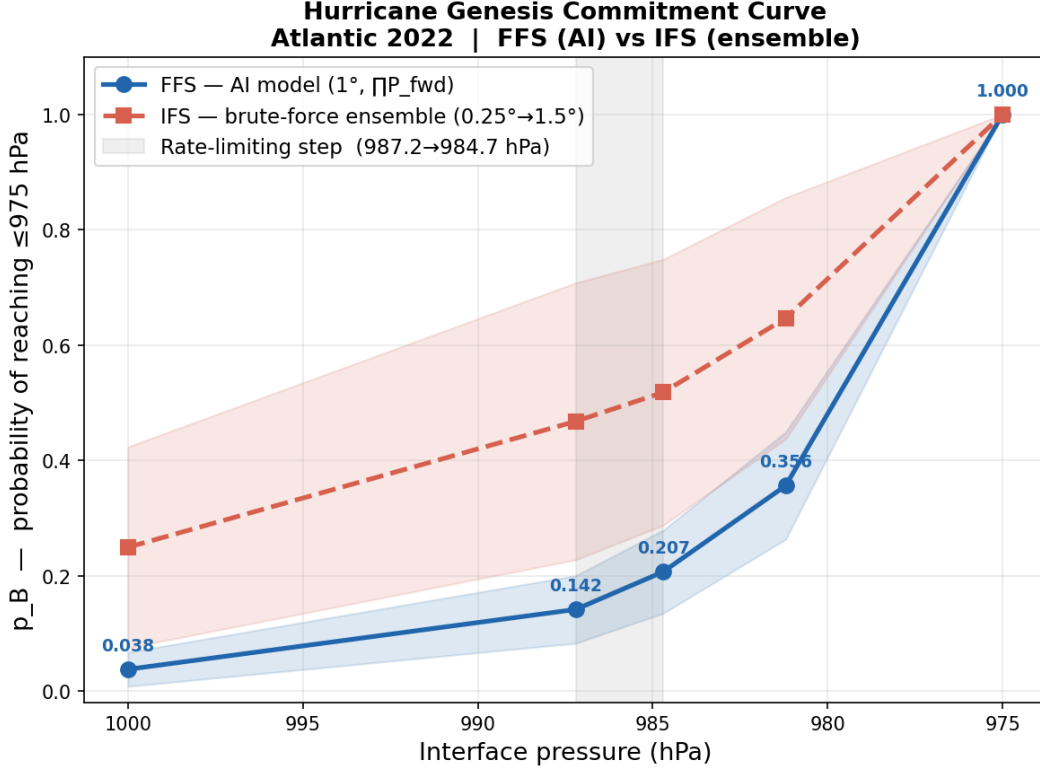


Figure 8: Genesis committor curve $p_B(\lambda_i)$ — the probability of eventually reaching state B (≤ 975 hPa) given arrival at interface λ_i — averaged across all 98 initial conditions. *Blue*: FFS AI model mean $\pm 1\sigma$ across ICs. *Orange*: IFS brute-force ensemble mean $\pm 1\sigma$ across ICs. The shaded bands show IC-to-IC variability; the annotated values are the all-season FFS means. The grey band highlights the rate-limiting step (largest single drop), which falls between the first and second shooting interfaces for the full 2022 Atlantic season. The FFS and IFS curves agree in the overall shape while differing in magnitude, consistent with the resolution and physics differences between the two systems.

3.5 Physics Along a Reactive Pathway

Figure 9 traces the storm-centred thermodynamic and kinematic evolution along a representative Earl reactive pathway (pathway 26) from λ_0 through state B . The vortex originates near 25°N and tracks northeastward to 39°N as it intensifies from 998 hPa to 967 hPa, consistent with typical late-summer Atlantic storm tracks in this environment. The persistent cyclonic vorticity (ζ_{850}) and warm-core anomaly (T'_{500}) at each interface confirm continuous vortex identity and warm-core structure through state B . These storm-centred panels reveal the physical mechanisms operating at each stage, connecting the abstract conditional probabilities in Table 1 to concrete dynamical processes.

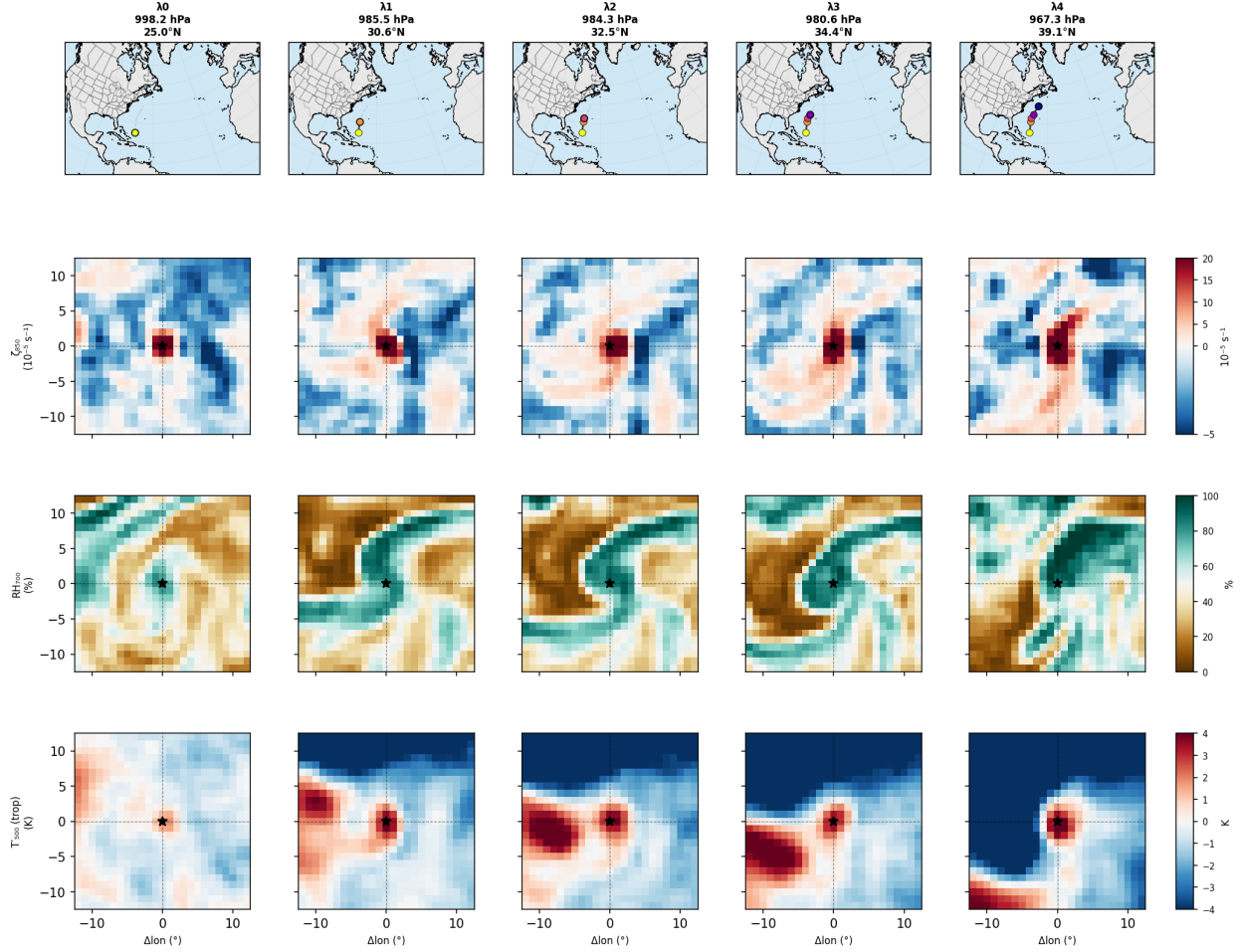


Figure 9: Storm-centred physics along a representative reactive pathway for the Earl IC (2022-09-02T00Z, pathway 26), λ_0 through λ_4 . Each column shows one interface crossing; column headers give the MSLP and storm latitude. The vortex tracks northeastward from 25°N at λ_0 to 39°N at λ_4 , consistent with typical late-summer Atlantic tracks in this environment. *Row 1*: 850-hPa relative vorticity ζ_{850} (10^{-5} s^{-1}). *Row 2*: 700-hPa relative humidity RH_{700} (%). *Row 3*: 500-hPa warm-core anomaly T'_{500} (K, relative to 0–30°N mean). The persistent cyclonic vorticity and warm-core anomaly at each interface confirm continuous vortex identity and warm-core structure through state *B*. Pathways where the warm core collapses at high latitudes are screened out by the CPS filter and not included in the genesis rate calculation.

3.6 Storm-Centred Physics Composites

Figure 10 shows storm-centred composites of vorticity, relative humidity, and warm-core anomaly, composited across all reactive pathways for each case-study IC. By averaging over all configurations that successfully reach state B , these composites reveal the mean thermodynamic and kinematic environment that supports genesis, smoothing out pathway-to-pathway variability to expose the underlying physical signal.

All three ICs share a common structural signature: a compact positive vorticity core at 850 hPa, a saturated inner core ($\text{RH}_{700} \gtrsim 70\%$), and a growing warm-core anomaly at 500 hPa that strengthens progressively from λ_0 through state B . The amplitude of T'_{500} at state B is comparable across the three cases despite their differing genesis rates, suggesting that once a vortex reaches state B it has traversed a physically similar intensification pathway regardless of the environmental background. The primary difference among the ICs appears in the earlier interfaces: the Earl IC ($k = 1.8 \times 10^{-3} \text{ day}^{-1}$) shows weaker vorticity organisation at λ_1 compared to the active-period IC, consistent with the lower P_1 value for that case (Table 1).

4 Discussion and Conclusions

4.1 Methodological Validation and Practical Considerations

The central validation result — FFS and direct-sampling rates agreeing within 1.03 ± 0.15 across 98 initial conditions — provides strong empirical support for the method. These are entirely independent estimates derived from the same simulation data: the FFS rate uses only the interface crossing counts and shooting probabilities, while the direct rate uses only the raw state- B arrival counts from flux trajectories. Their close agreement across three orders of magnitude of genesis probability confirms that (1) the interface spacing produces well-conditioned conditional probability estimates, (2) the ensemble sizes ($N = 2000$ per interface) are sufficient, and (3) the CPS contamination screen does not introduce systematic bias.

Enhancement factors (3–140 $\times$, geometric mean 14 $\times$, arithmetic mean 26 $\times$) scale inversely with genesis probability, as expected from FFS theory [Allen et al., 2009]. The log-normal character of the speedup distribution (geometric mean substantially below arithmetic mean) mirrors the log-normal FFS rate distribution itself and motivates reporting geometric rather than arithmetic means for speedup summaries. For the most active environments where direct sampling is feasible (speedup 3 $\times$), the two methods serve as mutual validation. For suppressed environments (speedup $\sim 100\times$), FFS is the only practical route to a quantitative genesis rate estimate.

The bottleneck analysis adds physical content beyond a scalar rate. For the Earl IC, the rate-limiting step at λ_1 (initial organization) indicates that the environment is resistant to producing organized convection but that once organized, the system is likely to reach hurricane strength. For the Ian precursor IC, the easy initial organization but hard final intensification ($P_4 = 0.102$) suggests an environment whose large-scale structure is favorable for genesis attempts but where

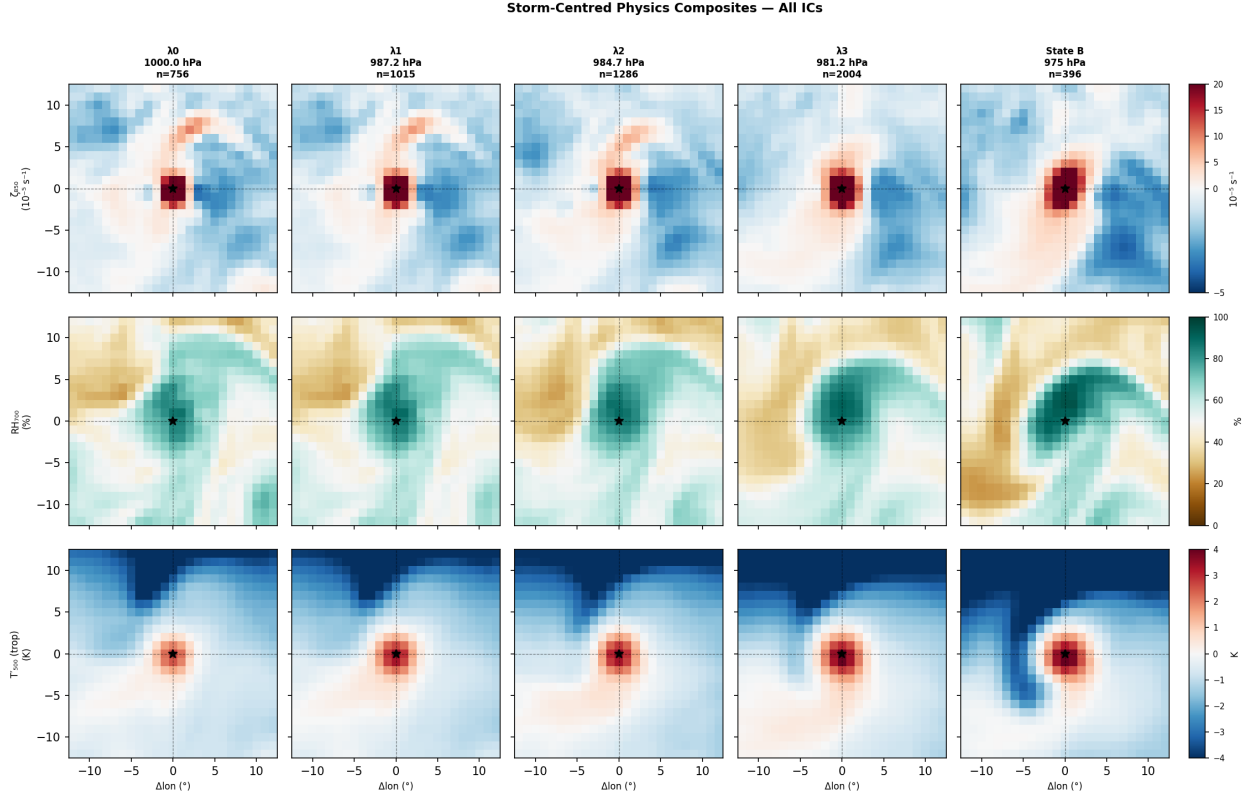


Figure 10: Storm-centred physics composites along reactive pathways, averaged over all reactive configurations for the three case-study ICs combined (all ICs panel). Columns show interfaces λ_0 through state B ; sample sizes n are given in each column header. *Row 1*: 850-hPa relative vorticity ζ_{850} (10^{-5} s^{-1}). *Row 2*: 700-hPa relative humidity RH_{700} (%). *Row 3*: 500-hPa warm-core anomaly T'_{500} (K, relative to $0\text{--}30^\circ\text{N}$ mean). The progressive intensification of the vortex core and warm anomaly from λ_0 to state B is robust across ICs, confirming that the FFS reactive pathways capture physically coherent tropical cyclogenesis rather than model noise.

the thermodynamic pathway to full hurricane development requires specific mesoscale alignment not yet present 24 hours before NHC designation.

The CPS filter warrants clarification. At 1° resolution, formal CPS parameters are not quantitatively validated against higher-resolution analyses, and our implementation omits the thermal asymmetry (B) parameter. The filter is best understood as a contamination screen rather than a rigorous warm-core classifier — rejecting obviously extratropical systems while accepting all ambiguous cases. Future implementations with finer-resolution models could apply a more principled multidimensional CPS criterion.

4.2 Limitations

This study analyzes a single hurricane season (2022). Interannual variability driven by ENSO, the AMO, and Saharan dust outbreaks is not represented. The SDL-WXFormer was trained on ERA5 (1979–2021) at 1° resolution, which smooths intense vortex cores and underrepresents Category 3+ hurricanes; state B is accordingly defined at 975 hPa rather than the intensities typical of major hurricane activity. Validation against physics-based models (WRF, IFS) at higher resolution is an important next step.

4.3 Conclusions

Forward Flux Sampling resolves Atlantic tropical cyclogenesis rates spanning three orders of magnitude from 98 atmospheric initial conditions in a fast neural weather emulator, with internal validation confirming FFS and direct-sampling rates agree within 3%. Computational enhancement of $3\text{--}140\times$ (geometric mean $14\times$) makes FFS practical across the full range of synoptic environments. Physically, the method reveals that the rate-limiting step in genesis varies by regime: initial tropical organization (P_1 bottleneck, as in Earl), uniformly elevated crossing probabilities with a terminal intensification barrier (P_4 bottleneck, as in the active-period IC), or a compound barrier at the final intensification stages ($P_3\text{--}P_4$, as in the Ian precursor). These distinct bottleneck signatures encode mechanistic information about genesis pathway dynamics that is invisible to scalar rate estimates from direct sampling.

The methodology extends directly to other atmospheric rare events (extreme heatwaves, sudden stratospheric warmings, blocking onset) and to physics-based models. The combination of neural emulators with rare event sampling opens a computationally tractable path to systematic characterization of low-probability, high-consequence atmospheric events across climatic regimes.

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Data Availability Statement

[Data availability statement required before submission. See comment above.]

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